

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Data Visualization with Power BI and Tableau

Practical – IX

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Class: TYIT	Batch: 2
Date of Assignment: 27-8-26	Date/Time of Submission: 27-8-26

Cleaning and Structuring Data for Analysis

Dataset: Custom Designed Students Dataset [csv files]

1) Sem1_Scores.csv

Student_ID	Full_Name	Dept_Info	Math	Physics	Chemistry	Programming	Attendance
S101	Aarav Sharma	CSE-01	85	90%	88	95	85 %
S102	priya VERMA	ECE-02		75%	80	85	90 %
S103	Rohan Gupta	CSE-01	78	82%		88	78 %
S104	Sneha Patel	IT-03	92	88%	90	96	95 %
S105	aditya Rao	MECH-04	45		50	60	65 %
S106	Ananya Singh	CSE-01	88	91%	85	92	88 %
S107	Rahul Nair	ECE-02		70%	75	80	72 %
S108	Kavya Reddy	IT-03	95	94%	92	98	98 %
S109	Vikram Joshi	CSE-01	67	60%	58	72	80 %
S110	Pooja Iyer	ECE-02	82	85%	80	89	91 %

2) Sem2_Scores. Csv

Student_ID	Full_Name	Dept_Info	Math	Physics	Chemistry	Programming	Attendance
S101	Aarav Sharma	CSE-01	88	92%	90	98	88 %
S102	Priya Verma	ECE-02	79	78%	82	88	92 %
S103	Rohan Gupta	CSE-01	82	85%	80	90	80 %
S104	Sneha Patel	IT-03	95	90%	94	98	96 %
S105	Aditya Rao	MECH-04	52	58%	55	65	70 %
S106	Ananya Singh	CSE-01	90	94%	88	95	90 %
S107	Rahul Nair	ECE-02	74	73%	78	82	75 %
S108	Kavya Reddy	IT-03	96	96%	95	99	99 %
S109	Vikram Joshi	CSE-01	70	65%	62	75	82 %
S110	Pooja Iyer	ECE-02	85	88%	84	92	94 %

3) Department_Master.csv

Dept_Code	Department_Name	HOD_Name	Dept_Target_Averag	Total_Dept_Capacit
CSE-01	Computer Science	Dr. Raman	85	120
ECE-02	Electronics & Comm	Dr. Iyer	78	90
IT-03	Information Tech	Dr. Sen	82	60
MECH-04	Mechanical Engg	Dr. Sharma	70	60

4) Student_Activities.csv

Student_ID	Club_Membership	Projects_Completed	Certifications
S101	Coding Club	3	2
S102	Robotics Club	2	1
S103	Coding Club	2	1
S104	Debate Society	4	3
S105	Sports Club	1	0
S106	Coding Club	3	2
S107	Robotics Club	2	1
S108	Debate Society	4	3
S109	Sports Club	1	1
S110	Cultural Club	2	2

A: Identify and Handle Missing or Inconsistent Data

Step 1: Load the Dataset

1. Open Tableau Desktop.
2. Under the Connect pane on the left, click Text file.
3. Select and open Sem1_Scores.csv.
4. In the Data Source preview table:
 - o Observe missing entries (Null) in Math, Physics, and Chemistry.

Student ID	Full Name	Dept Info	Math	Physics	Chemistry	Programming	Attendance
S101	Aarav Sharma	CSE-01	85	0.900000	88	95	85 %
S102	priya VERMA	ECE-02	null	0.750000	80	85	90 %
S103	Rohan Gupta	CSE-01	78	0.820000	null	88	78 %
S104	Sneha Patel	IT-03	92	0.880000	90	96	95 %
S105	aditya Rao	MECH-04	45	null	50	60	65 %
S106	Ananya Singh	CSE-01	88	0.910000	85	92	88 %
S107	Rahul Nair	ECE-02	null	0.700000	75	80	72 %

- o Note that Physics and Attendance are classified as text (Abc) due to the trailing % characters.

Step 2: Clean Inconsistent String Fields (Convert Text Percentages to Numeric Measures)

1. Navigate to Sheet 1 (bottom-left tab).
2. Right-click an empty space in the Data Pane (left sidebar) --> select Create Calculated Field....
3. Create the cleaned measure for Physics:
 - o Field Name: Physics_Clean
 - o Tableau Formula:


```
Code snippet : FLOAT(REPLACE(STR([Physics]), '%', ''))
```
 - o Click OK.

Physics_Clean

`FLOAT(REPLACE(STR([Physics]), '%', ''))`

The calculation is valid.

Apply OK

4. Create the cleaned measure for Attendance:
 - o Right-click the Data Pane --> Create Calculated Field....

- o Field Name: Attendance_Clean
- o Tableau Formula:

Code snippet : `FLOAT(TRIM(REPLACE(STR([Attendance]), '%', '')))`

- o Click OK.

Attendance_Clean

`FLOAT (TRIM (REPLACE (STR ([Attendance]), '%', '')))`

The calculation is valid.
Apply
OK

Step 3: Handle Missing (Null) Numeric Values via Imputation

1. Handle missing entries in Math:

- o Right-click the Data Pane --> Create Calculated Field....
- o Field Name: Math_Clean
- o Tableau Formula:

Code snippet : `IFNULL([Math], 0)`
- o Click OK.

Math_Clean

`IFNULL ([Math], 0)`

The calculation is valid.
Apply
OK

2. Handle missing entries in Chemistry:

- o Right-click the Data Pane --> Create Calculated Field....
- o Field Name: Chemistry_Clean
- o Tableau Formula:

Code snippet : `IFNULL([Chemistry], 0)`

Chemistry_Clean

`IFNULL ([Chemistry], 0)`

The calculation is valid.
Apply
OK

Step 4: Impute Missing Values in Physics (Optional Handling for Student S105)

- Student S105 contains a Null value in Physics. Combine string stripping with null handling:
 - Edit Physics_Clean (right-click Physics_Clean --> Edit...):
 - Updated Tableau Formula:

Code snippet

`IFNULL(FLOAT(REPLACE(STR([Physics]), '%', '')), 0)`
 - Click OK.



B: Convert Wide Data into Tall Format

Step 1: Open the Data Source Grid

- In Tableau, navigate to the Data Source tab at the bottom-left corner.
- Locate the four horizontal subject score columns:
 - Math
 - Physics
 - Chemistry
 - Programming

Fields

Type	Field Name	Physical Table	Remote...
Abc	Attendance ▼	Sem1_Scores.csv	Attenda...
=#	Physics_Clean ▼	Calculation	Calculati...
=#	Attendance_Clean ▼	Calculation	Calculati...
=#	Math_Clean ▼	Calculation	Calculati...
=#	Chemistry_Clean ▼	Calculation	Calculati...

Step 2: Multi-Select and Pivot Columns

- Click the header of the Math column.
- Hold down Ctrl (or Cmd on Mac) and click Physics, Chemistry, and Programming so all four columns are highlighted in blue
- Right-click on any of the highlighted column headers.
- Select Pivot.

1. Double-click the header named **Pivot Field Names** and type: **Subject**
2. Double-click the header named **Pivot Field Values** and type: **Score**

Step 4: Clean and Format the Tall Score Field Because raw Physics contained % symbols, the new combined Score column defaults to text (Abc). Convert it to a numeric measure:

- Sem1

✓ Number (decimal)

Number (whole)

Date & Time

Date

String

Spatial

Boolean

Default

Geographic Role

fields 30 rows

30

→

rows

⚙

▼

Dept Info	Math	Attendance	Subject	Pivot Field Values	Score
CSE-01	85	null	Chemistry	88.0000	n
CSE-01	85	null	Physics	0.9000	n
CSE-01	85	null	Programm...	95.0000	n
ECE-02	null	null	Chemistry	80.0000	n
ECE-02	null	null	Physics	0.7500	n
ECE-02	null	null	Programm...	85.0000	n

- 5

Code snippet :

```
IFNULL(FLOAT(REPLACE(STR([Score]), '%', '')), 0)
```

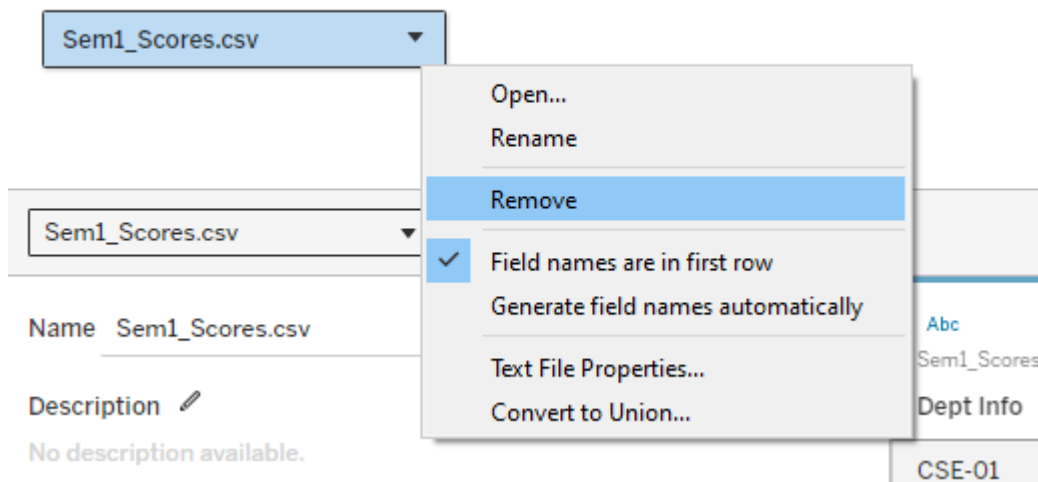
Clean Score



C: Perform Union of Multiple Files

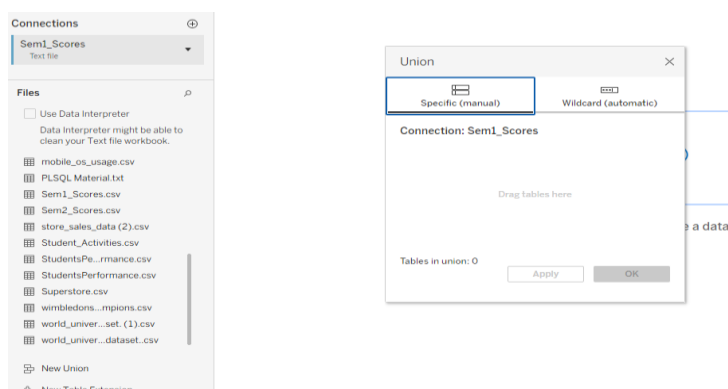
Step 1: Open the Data Source Canvas

1. Open Tableau Desktop and go to the Data Source tab in the bottom-left corner.
2. If any single table is already on the canvas, right-click it and select Remove to start with a clean canvas.



Step 2: Create a Multi-File Union

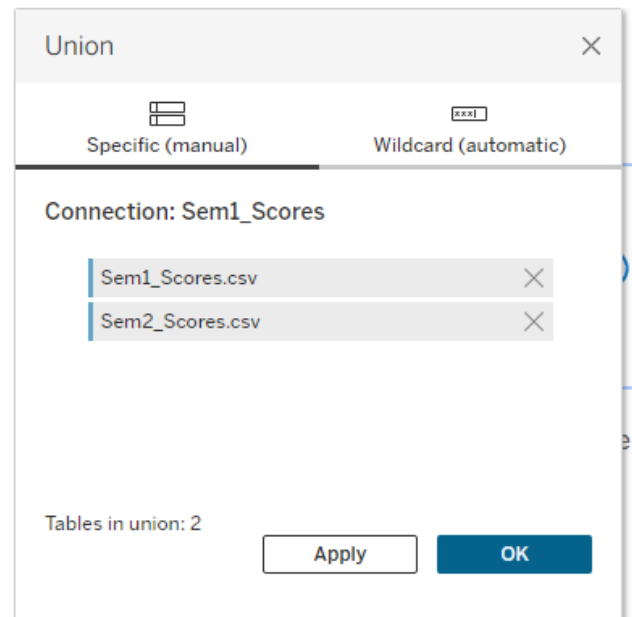
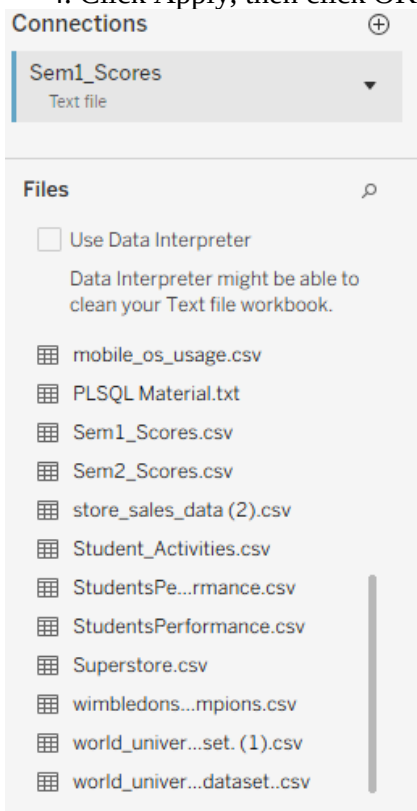
1. Look at the left sidebar under the Files / Connections pane.
2. Drag the New Union option onto the empty canvas area.



3. In the pop-up Union dialog box:

- o Drag Sem1_Scores.csv from the left panel into the box.
- o Drag Sem2_Scores.csv from the left panel into the same box.
- o Verify that both filenames are listed in the Union box.

4. Click Apply, then click OK.



5. The canvas table icon will now display with a plus symbol (+), representing the consolidated union.

Step 3: Verify the Merged Data Grid

1. Review the data grid at the bottom of the screen.
2. Confirm the row count: the total number of rows increases from 10 to 20 rows (10 records for Semester 1 and 10 records for Semester 2).
3. Scroll to the far right of the grid to verify the generated system column:
 - o Table Name: Displays "Sem1_Scores.csv" for the first 10 rows and "Sem2_Scores.csv" for the remaining 10 rows

Sem1_Scores

Connection: ☒ Live ☐ Extract Filters: 0 | Add

Union

Union 14 fields 20 rows 20 rows

Name	Union	Union	Union	Calculation	Calculation
Description	Programmer	Attendance	Table Name	Clean_Score	Score
92	98	98 %	Sem1_Scores.csv	null	0.940000
58	72	80 %	Sem1_Scores.csv	null	0.600000
80	89	91 %	Sem1_Scores.csv	null	0.850000
90	98	88 %	Sem2_Scores.c...	null	0.920000
82	88	92 %	Sem2_Scores.c...	null	0.780000
80	90	80 %	Sem2_Scores.c...	null	0.850000
0.4	0.8	0.8 %	Sem2_Scores.c...	null	0.000000

Type	Field Name	Physical Table	Remote Field...
Abc	Student_ID	Union	Student_ID
Abc	Full_Name	Union	Full_Name
Abc	Dept_Info	Union	Dept_Info

Step 4: Create a Clean Semester Identifier Dimension

1. Click on Sheet 1 at the bottom-left.
2. In the Data Pane (left sidebar), right-click in an empty space $\diamond$ select Create Calculated Field....
3. Set the Field Name to: Semester 4. Enter the following formula:

Code snippet :

```
IF CONTAINS([Table Name], "Sem1") THEN "Semester 1"
ELSEIF CONTAINS([Table Name], "Sem2") THEN "Semester 2"
ELSE "Other"
END
```

Clean Score



5. Click OK.

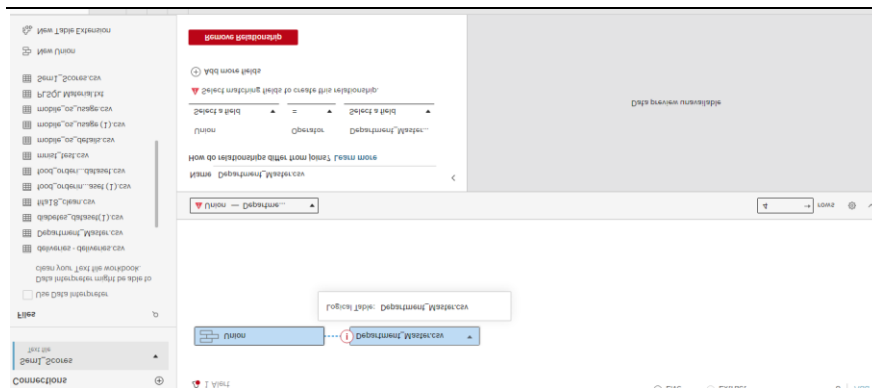
D: Execute Cross-Database Joins

Step 1: Stay on the Top Canvas

- Leave your Union table box as it is on the canvas.

Step 2: Add and Link Department_Master.csv

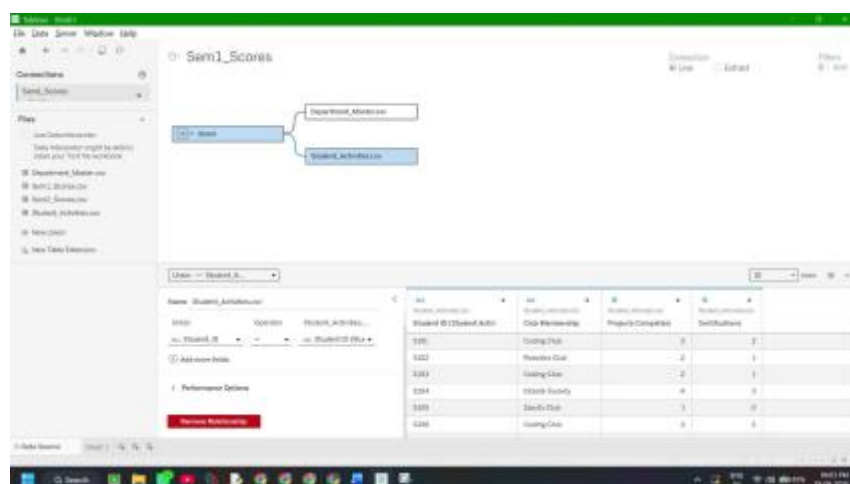
1. In the left sidebar under Files / Connections, find Department_Master.csv.
2. Drag Department_Master.csv onto the canvas and drop it to the right of your Unionbox.
3. A flexible line (relationship / join noodle) will connect the two boxes.



4. If the relationship dialog opens (or click the relationship line):
 - o Left Table Field (Union): Select Dept_Info
 - o Right Table Field (Department_Master): Select Dept_Code
 - o Operator: =

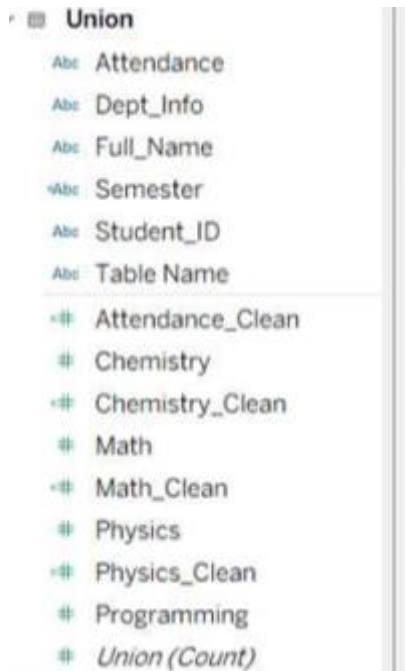
1. From the left sidebar, drag `Student_Activities.csv` onto the canvas.

o Left Table Field (Union): Select Student_ID o Right Table Field (Student_Activities): Select Student_ID
o Operator: =



Step 4: Verify in Sheet 1

1. Click on Sheet 1 at the bottom-left.
2. In the Data Pane on the far left, you will now see all three tables organized with their respective fields:
 - o Union: Student_ID, Full_Name, Dept_Info, Math, Physics, Chemistry, Programming, Attendance, Table Name
 - o Department_Master: Dept_Code, Department_Name, HOD_Name, Dept_Target_Average, Total_Dept_Capacity
 - o Student_Activities: Club_Membership, Projects_Completed, Certifications



E: Restructure Data for Better Visualization

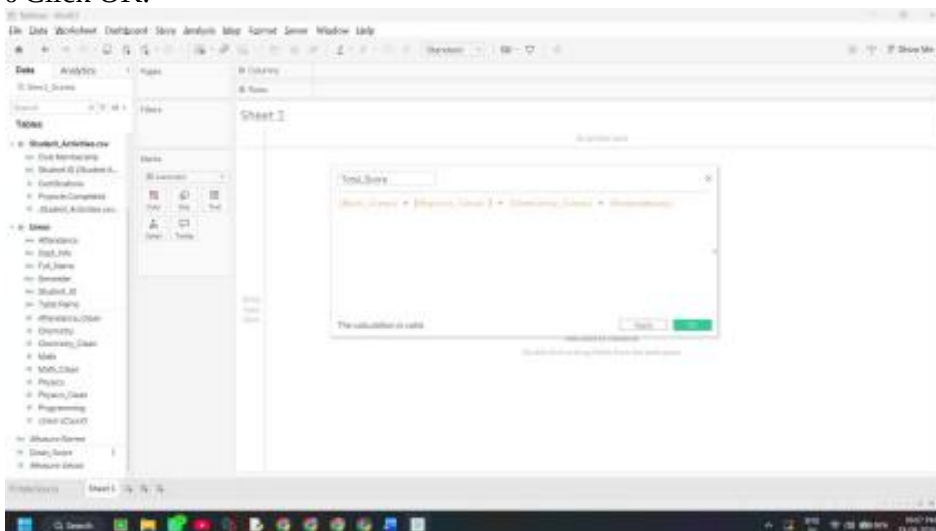
Step 1: Calculate Total Marks and Scaled Percentage

1. Navigate to Sheet 1 in Tableau.
2. In the Data Pane on the left, right-click an empty space --> select Create Calculated Field....
3. Create the student's total marks field:
 - o Field Name: Total_Score
 - o Tableau Formula:

Code snippet :

$[Math_Clean] + [Physics_Clean] + [Chemistry_Clean] + [Programming]$

- o Click OK.

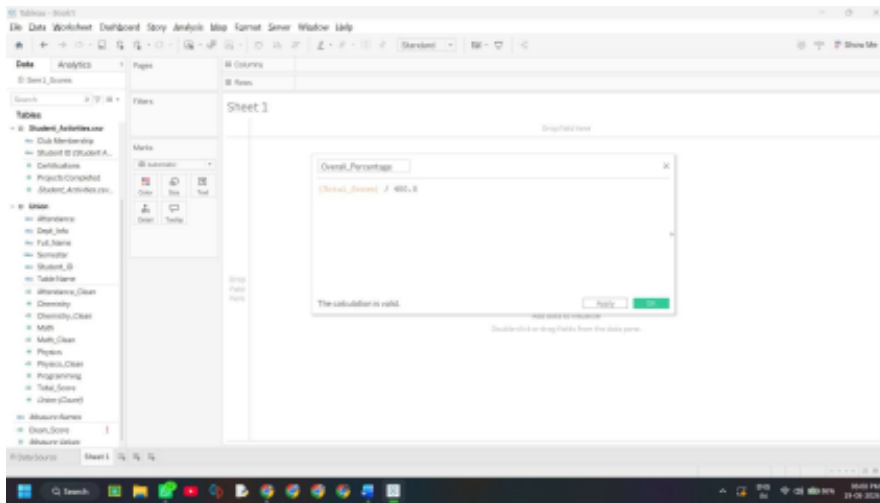


4. Create the scaled overall percentage field:

- o Right-click in the Data Pane --> select Create Calculated Field....
- o Field Name: Overall_Percentage
- o Tableau Formula:

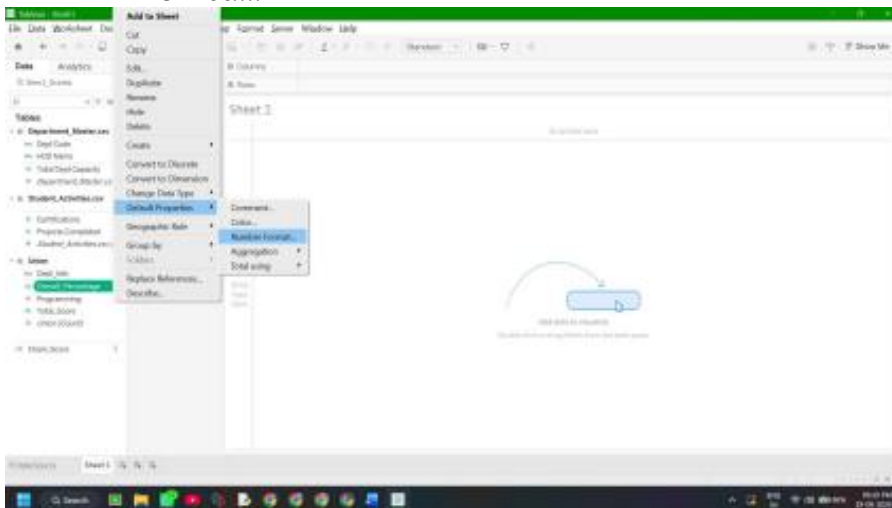
Code snippet : $[Total_Score] / 400.0$

- o Click OK.

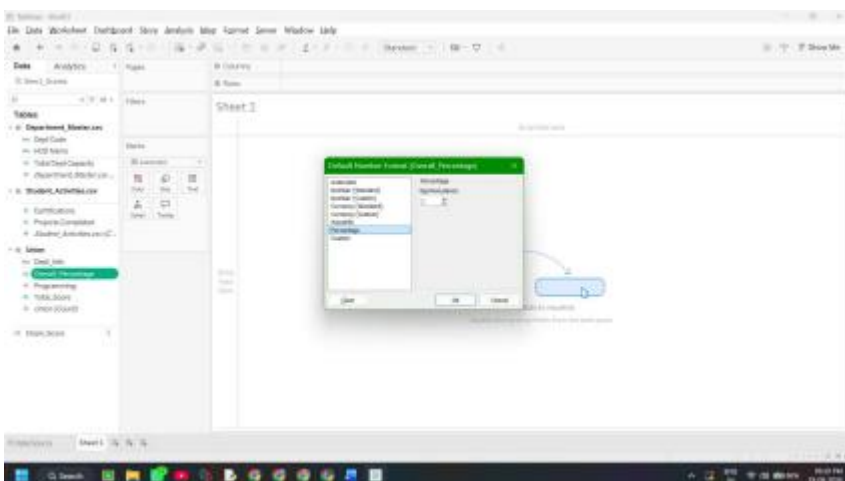


5. Set the default percentage display formatting:

- o In the Data Pane, right-click Overall_Percentage --> Default Properties --> Number Format....



- o Select Percentage and set decimal places to 1 (e.g., 85.5%).
- o Click OK.



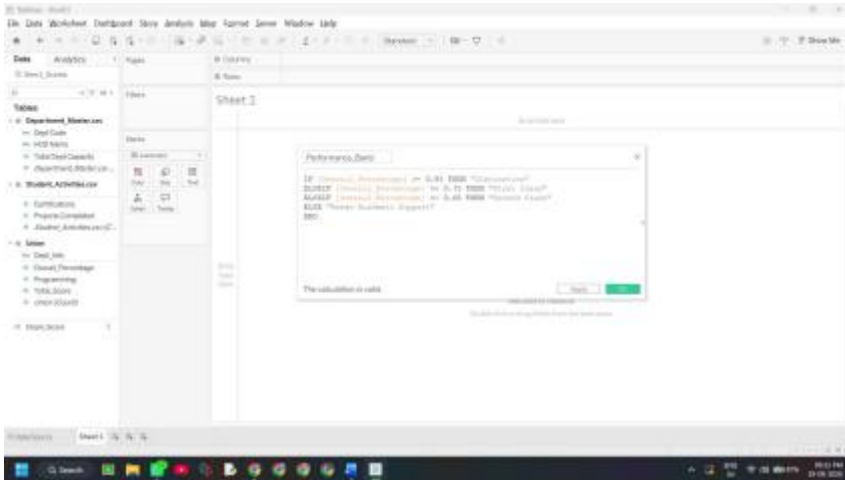
Step 2: Create Categorical Performance Bands (Grading Tiers)

1. Right-click in the Data Pane -->select Create Calculated Field....
2. Create the academic grading tier field:
 - o Field Name: Performance_Band o Tableau Formula:

Code snippet

```
IF [Overall_Percentage] >= 0.85 THEN "Distinction"
ELSEIF [Overall_Percentage] >= 0.75 THEN "First Class"
ELSEIF [Overall_Percentage] >= 0.60 THEN "Second Class"
ELSE "Needs Academic Support" END
```

- o Click OK.



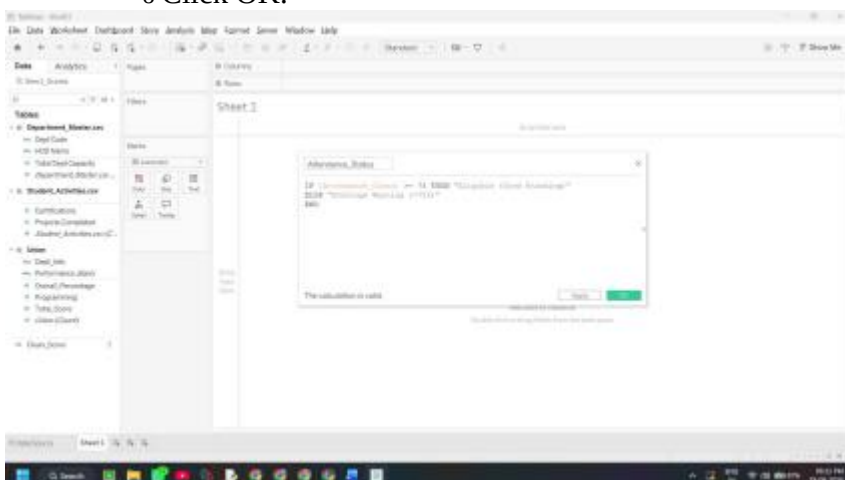
Step 3: Create an Attendance Health Flag

1. Right-click in the Data Pane -->select Create Calculated Field....
2. Create the attendance compliance indicator:
 - o Field Name: Attendance_Status o Tableau Formula:

Code snippet

```
IF [Attendance_Clean] >= 75 THEN "Eligible (Good Standing)" ELSE "Shortage Warning (<75%)" END
```

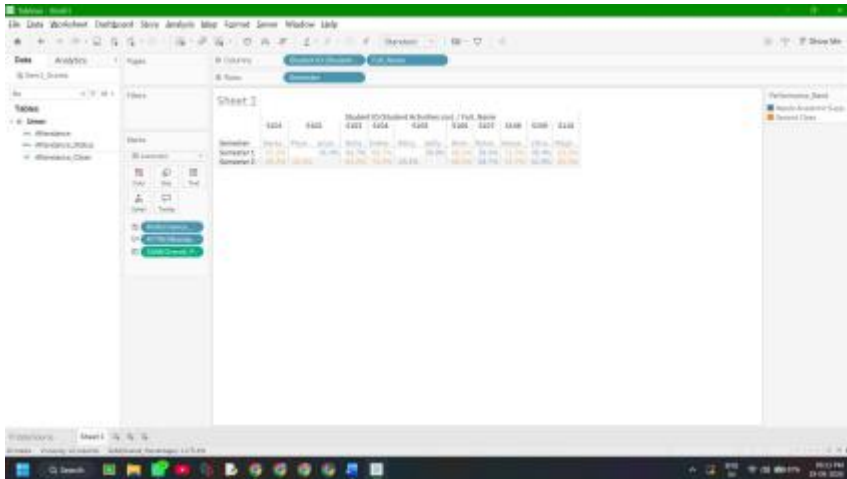
- o Click OK.



Step 4: Build the Student Performance Scorecard View

1. Drag Student_ID and Full_Name onto the Rows shelf.
2. Drag Semester onto the Columns shelf.
3. Drag Overall_Percentage onto the Text tile on the Marks card.

4. Drag Performance_Band onto the Color tile on the Marks card.
5. Drag Attendance_Status onto the Tooltip (or Detail) tile on the Marks card.



F: Handle Different Levels of Detail in Datasets

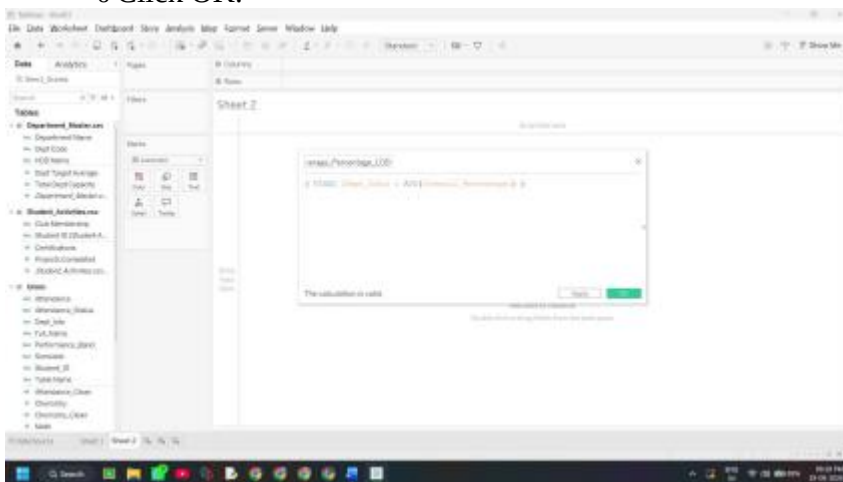
Step 1: Compute Department-Level Benchmark using {FIXED} LOD

1. In Tableau, navigate to Sheet 1 (or create a new worksheet named Sheet 2).
2. In the Data Pane on the left, right-click an empty space --> select Create Calculated Field....
3. Create the department-wide benchmark field:
 - o Field Name: Dept_Average_Percentage_LOD
 - o Tableau Formula:

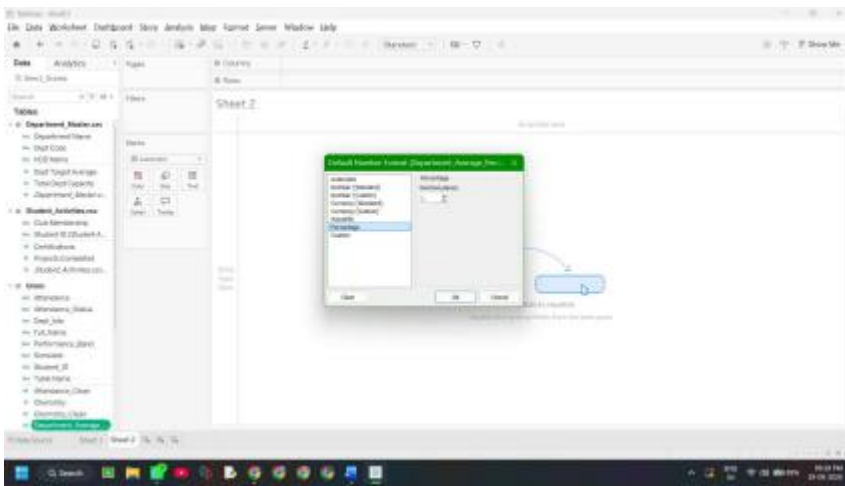
Code snippet

```
{ FIXED [Dept_Info] : AVG([Overall_Percentage]) }
```

- o Click OK.



4. Set default formatting:
 - o Right-click Dept_Average_Percentage_LOD --> Default Properties --> Number Format... --> select Percentage (1 decimal place) --> OK.



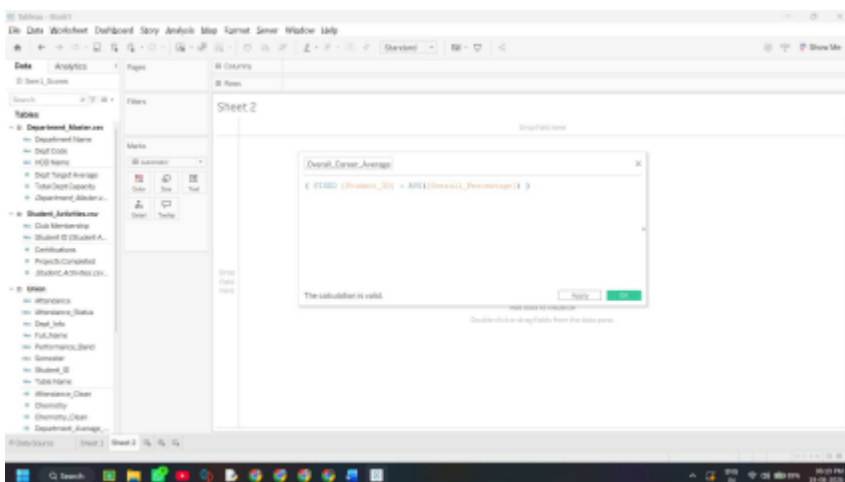
Step 2: Compute Student Multi-Semester Average using {FIXED} LOD

1. Right-click in the Data Pane --> select Create Calculated Field....
2. Create the cumulative student benchmark across all semesters:
 - o Field Name: Student_Overall_Career_Average
 - o Tableau Formula:

Code snippet

```
{ FIXED [Student_ID] : AVG([Overall_Percentage]) }
```

- o Click OK.



3. Set its Number Format to Percentage (1 decimal place, Just like we did in previous step).

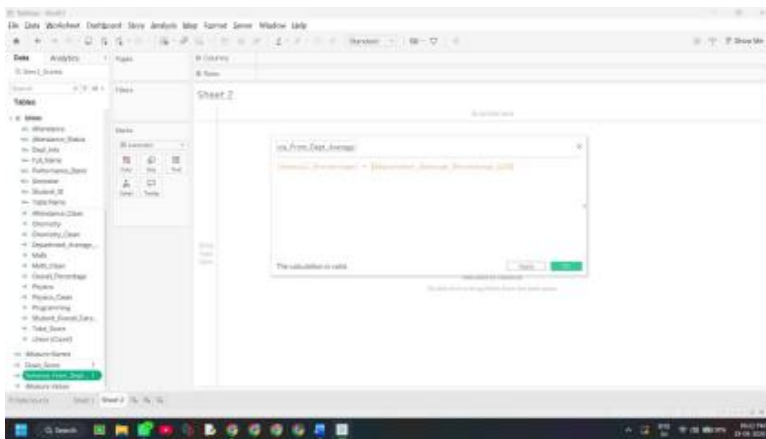
Step 3: Calculate Performance Variance (Student vs. Department Benchmark)

1. Right-click in the Data Pane --> select Create Calculated Field....
2. Create the relative difference metric:
 - o Field Name: Variance_From_Dept_Average
 - o Tableau Formula:

Code snippet

```
[Overall_Percentage] - [Dept_Average_Percentage_LOD]
```

- o Click OK.



3. Set its Number Format to Percentage (1 decimal place).

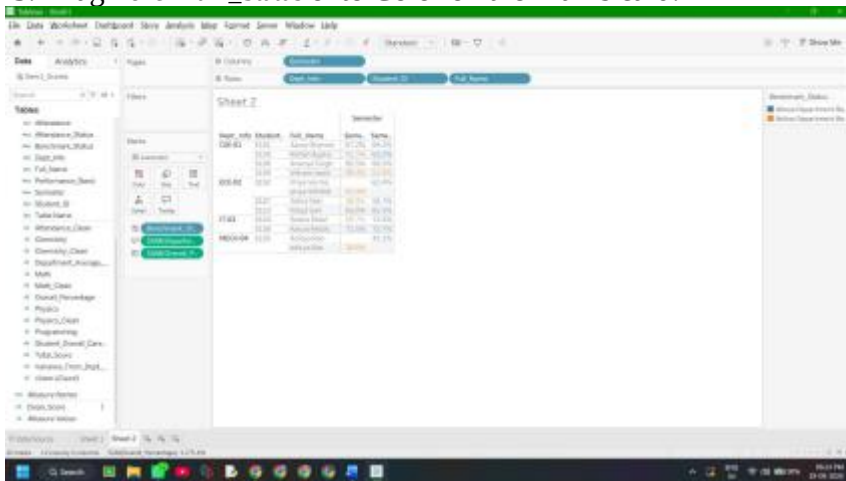
Step 4: Create a Benchmark Indicator Dimension

1. Right-click in the Data Pane --> select Create Calculated Field....
2. Create an automated comparison label:
 - o Field Name: Benchmark_Status
 - o Tableau Formula:


```
Code snippet
IF [Variance_From_Dept_Average] >= 0 THEN "Above Department Benchmark"
ELSE "Below Department Benchmark" END
```
 - o Click OK.

Step 5: Build the LOD Benchmark Visualization

1. Drag Dept_Info, Student_ID, and Full_Name to the Rows shelf.
2. Drag Semester to the Columns shelf.
3. Drag Overall_Percentage onto Text on the Marks card.
4. Drag Dept_Average_Percentage_LOD onto Tooltip on the Marks card.
5. Drag Benchmark_Status onto Color on the Marks card.



G: Apply Advanced Data Cleaning Techniques

Step 1: Clean Leading/Trailing Whitespace and Inconsistent Name Casing

1. Navigate to Sheet 1 (or a new worksheet).
2. Right-click an empty space in the Data Pane --> select Create Calculated Field....
3. Create the trimmed and standardized student name field:
 - o Field Name: Full_Name_Clean
 - o Tableau Formula:

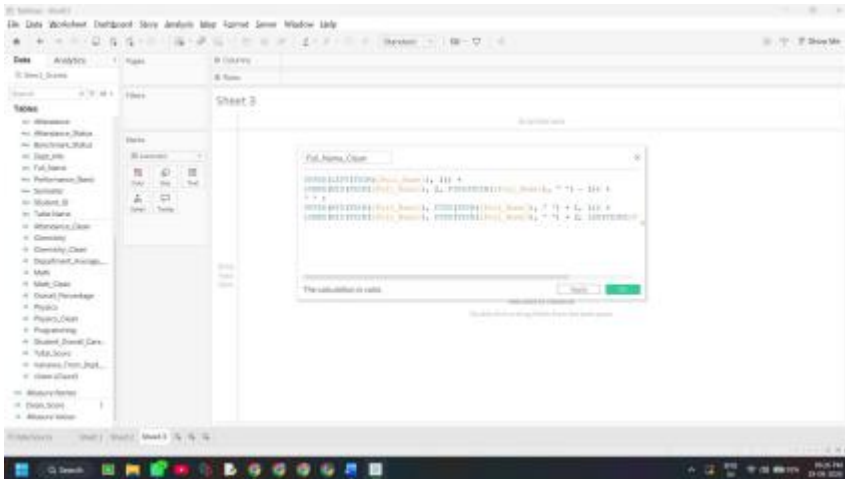
Code snippet :

```
UPPER(LEFT(TRIM([Full_Name]), 1)) +
LOWER(MID(TRIM([Full_Name]), 2, FIND(TRIM([Full_Name]), " ") - 1)) +
" " +
UPPER(MID(TRIM([Full_Name]), FIND(TRIM([Full_Name]), " ") + 1, 1)) +
LOWER(MID(TRIM([Full_Name]), FIND(TRIM([Full_Name]), " ") +
2, LEN(TRIM([Full_Name]))))
```

o (Alternatively, for standard trim-only cleaning):

Code snippet : TRIM([Full_Name])

o Click OK.



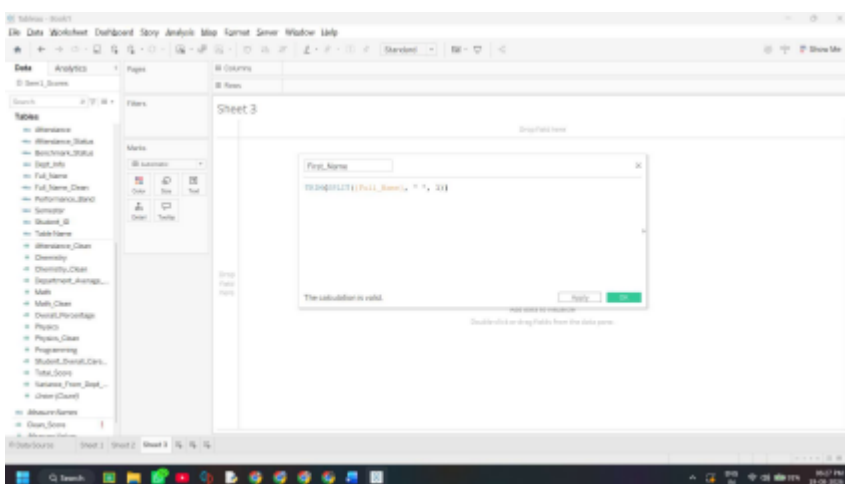
Step 2: Parse First Name and Last Name using SPLIT()

1. Extract the First Name:

- o Right-click in the Data Pane --> select Create Calculated Field....
- o Field Name: First_Name
- o Tableau Formula:

Code snippet : TRIM(SPLIT([Full_Name], " ", 1))

o Click OK.

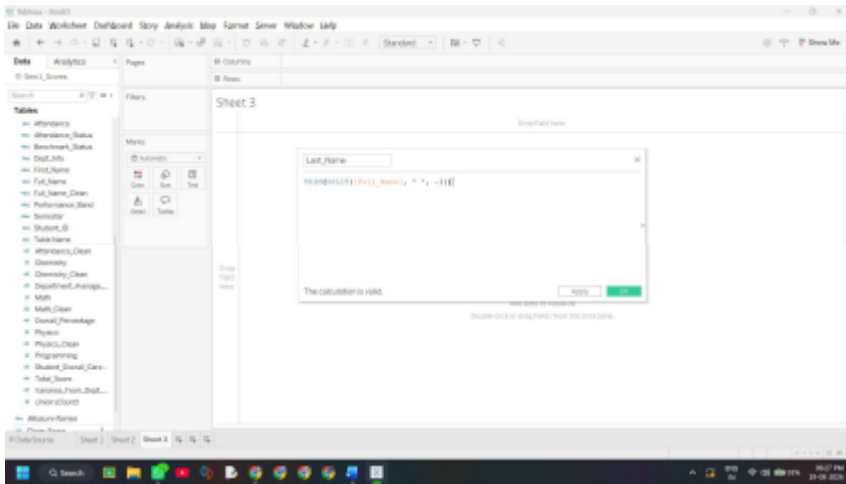


2. Extract the Last Name:

- o Right-click in the Data Pane --> select Create Calculated Field....
- o Field Name: Last_Name
- o Tableau Formula:

Code snippet : `TRIM(SPLIT([Full_Name], " ", -1`

o Click OK.



Step 3: Split Composite Department Codes (Dept_Info)

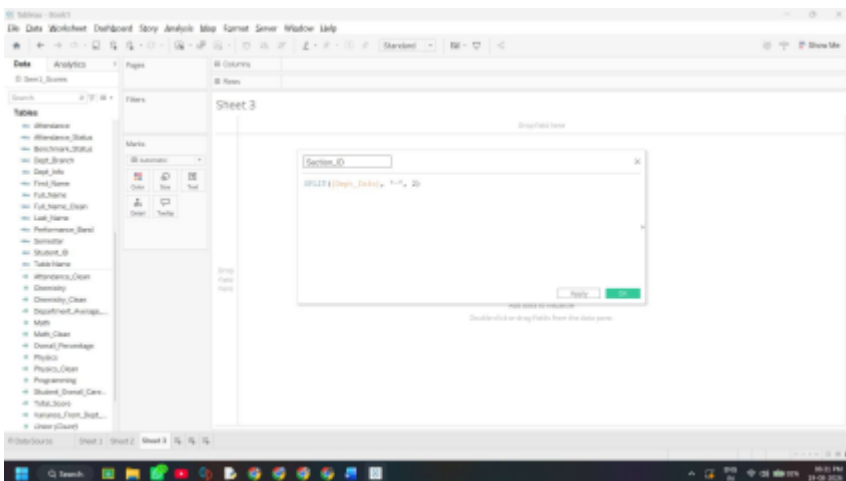
The column Dept_Info contains combined strings like CSE-01, ECE-02, and MECH-04. Separate them into distinct dimensional attributes:

1. Extract the Department Acronym:

- o Right-click in the Data Pane --> select Create Calculated Field....
- o Field Name: Dept_Branch
- o Tableau Formula:
Code snippet
`SPLIT([Dept_Info], "-", 1)`
- o Click OK.

2. Extract the Batch / Section ID:

- o Right-click in the Data Pane --> select Create Calculated Field....
- o Field Name: Section_ID
- o Tableau Formula:
Code snippet
`SPLIT([Dept_Info], "-", 2)`
- o Click OK.



Step 4: Create a Clean Composite Student Label

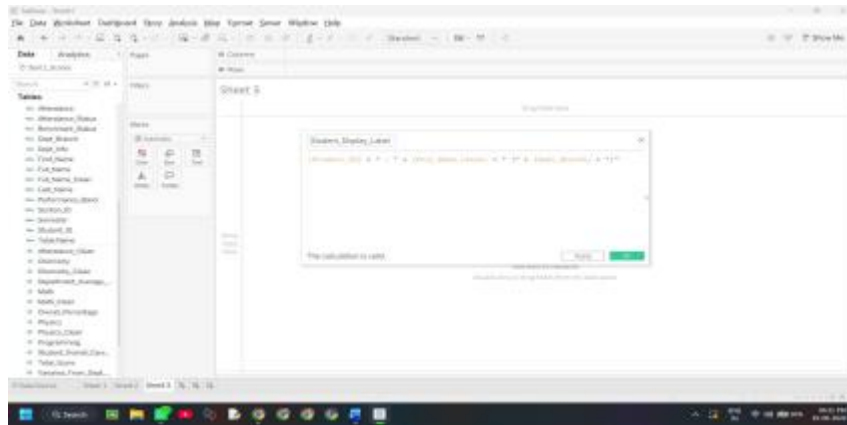
Combine cleaned names, student IDs, and department codes for clean dashboard titles and search filters:

1. Right-click in the Data Pane --> select Create Calculated Field....
2. Field Name: Student_Display_Label
3. Tableau Formula:

Code snippet :

`[Student_ID] + " - " + [Full_Name_Clean] + " (" + [Dept_Branch] + ")"`

4. Click OK.



Step 5: Verify in the Final Dashboard View

1. Place Student_Display_Label on the Rows shelf.
2. Place Semester on the Columns shelf.
3. Place Overall_Percentage on Text.
4. Place Performance_Band on Color.
5. Place Dept_Branch and Section_ID in the Filters shelf. (While Filtering You Can select any of the value you want)

